Q U A N T U M L A T T I C E M O D E L

Geometric Routing Admittance and Hard-Core Reflection

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Abstract

This paper develops the geometric and dynamical completion layer of the Quantum Lattice Model (QLM) following the emergence of Lorentz symmetry from invariant lattice transport. First, a dimensionless geometric routing admittance $\mathcal{Y}$ is derived as the normalized effective admittance ratio determined purely by boundary geometry, establishing a general routing-suppression principle for confined phase–action transport. Stable closure radii follow directly from this geometric suppression.

Second, the universal Planck energy-density cap is promoted from a static bound to a dynamical mechanism via minimal impedance blow-up, enforcing a finite saturated core during gravitational collapse. Collapse therefore terminates at a radius scaling as $M^{1/3}$ while the exterior solution remains exactly Schwarzschild, since the vacuum field depends only on the total enclosed mass.

Gravity is interpreted as asymptotic throttling of local action transport, while local lattice invariants remain unchanged. No new primitives or free parameters are introduced.

1 Introduction

As established in [11], Lorentz kinematics and the invariant bridge $E/p = c$ emerge directly from the per-radian Planck primitives $\{\hbar, \ell_{\mathrm{P}}, t_{\mathrm{P}}\}$ and invariant proper-tick accumulation. With kinematic structure fixed by lattice transport, the next structural layer concerns geometric routing of phase–action flow and the dynamical completion of gravitational collapse.

This paper develops that completion in two steps:

1. A dimensionless geometric routing admittance $\mathcal{Y}$ is derived as the normalized effective admittance ratio determined purely by boundary geometry, providing a general routing-suppression principle.

2. The universal Planck energy-density cap is promoted to a dynamical saturation mechanism via impedance blow-up, enforcing a finite interior core during gravitational collapse.

No new primitives are introduced. All results follow algebraically from impedance normalization, closure geometry, and the Planck density cap already implied by the QLM lattice rules [9, 10].

2 Canonical Primitives and Impedance Normalization

QLM adopts per-radian primitives

$$\{\hbar,\ \ell_{\mathrm{P}},\ t_{\mathrm{P}}\}, \qquad c = \frac{\ell_{\mathrm{P}}}{t_{\mathrm{P}}}. \tag{1}$$

Impedance in QLM is defined operationally as the ratio of transverse action flux to phase-gradient response in the null limit. Because action transport is fundamental, impedance normalization must be expressed purely in terms of lattice primitives.

In the vacuum (unsuppressed routing), transverse action flux propagates at the invariant null speed c without geometric confinement. The corresponding reference impedance scale is therefore set by the ratio of action transport per unit area to phase-gradient rate under full availability conditions.

For routing comparisons it is convenient to introduce a geometrically unwrapped normalization that removes the 4π spherical wrapping factor associated with isotropic emission. This isolates purely geometric suppression effects.

Define the Planck-normalized impedance and admittance:

$$Z_{\mathrm{P}} \equiv \frac{Z_0}{4\pi}, \qquad Y_{\mathrm{P}} \equiv \frac{1}{Z_{\mathrm{P}}}. \tag{2}$$

Here Z_0 denotes the vacuum wave impedance (unsuppressed vacuum routing impedance) determined by invariant lattice transport, and Y_{P} sets the reference (full-availability) routing baseline of the lattice.

Effective routing admittances will be defined as dimensionless admittance ratios relative to this Planck-normalized baseline, so that geometric suppression factors depend only on boundary geometry and not on microscopic constants.

3 Derived Routing Admittance

3.1 Definition

Let Y_{eff} denote the effective admittance of an allowed closed routing path determined solely by boundary geometry and available phase–transport channels. Here admittance refers to normalized phase–action transport capacity, not electromagnetic material response. Boundary geometry modifies effective routing multiplicity without altering primitive lattice constants; suppression arises from geometric restriction of admissible transport channels.

Define the dimensionless geometric routing admittance

$$\boxed{\mathcal{Y} \equiv \frac{Y_{\mathrm{eff}}}{Y_{\mathrm{P}}} = \frac{Z_{\mathrm{P}}}{Z_{\mathrm{eff}}}.} \tag{3}$$

$\mathcal{Y}$ is dimensionless and purely geometric. It is not a kinematic velocity parameter and does not represent mass; it encodes suppression of admissible routing channels relative to full lattice availability.

Throughout this section, r denotes a generic confinement radius. Specific closure radii will be indicated by subscripts.

3.2 Loop Momentum Law

Closed confinement in QLM corresponds to coherent phase closure around a spatial loop of radius r. Because each radian of phase carries reduced action $\hbar$, a spatial phase gradient generates momentum according to

$$\boxed{p = \hbar k,} \tag{4}$$

where k is the spatial phase rate.

For a closed loop, phase coherence requires the closure condition

$$\oint k\, dl = 2\pi n, \tag{5}$$

with integer winding number n. For a circular loop of radius r this gives

$$k(2\pi r) = 2\pi n \quad \Rightarrow \quad k = \frac{n}{r}. \tag{6}$$

The dominant confinement mode corresponds to the lowest nonzero winding number $n = 1$, yielding

$$\boxed{p_{\text{loop}}(r) = \frac{\hbar}{r}.} \tag{7}$$

This is the confinement closure relation derived in [9], where mass is interpreted as closed momentum inventory normalized by the invariant transport speed.

If routing is suppressed by the geometric admittance $\mathcal{Y}$, the effective confined momentum becomes

$$\boxed{p(r) = \mathcal{Y}\frac{\hbar}{r}, \qquad E(r) = p(r)c.} \tag{8}$$

Thus closure radii follow directly from routing geometry: suppression of admissible transport channels increases the required radius for a fixed mass scale.

3.3 Mass-Closure Baseline

For full routing ($\mathcal{Y} = 1$), the confined momentum inventory is identified with relativistic rest momentum,

$$p = mc.$$

Equating this with (7) fixes the minimal closure radius

$$\boxed{r_{\text{mass}}(m) = \frac{\hbar}{mc}.} \tag{9}$$

This is the reduced Compton scale [7] and represents the minimal closure radius under unrestricted routing.

3.4 Coulomb-Bound Routing

Empirically, electromagnetic binding in hydrogenic systems is governed by the dimensionless fine-structure constant [8]

$$\boxed{\alpha \approx 7.297\,352\,56 \times 10^{-3} \approx \frac{1}{137.036}.} \tag{10}$$

Within QLM, α is interpreted as a routing-suppression factor: it measures the fraction of full lattice routing capacity available under Coulomb confinement.

In Coulomb-bound systems, geometric and interaction constraints restrict admissible phase–action transport channels to a fraction

$$\mathcal{Y} = \alpha, \tag{11}$$

relative to full availability.

Substituting this routing suppression into the loop momentum law (8) yields the confinement radius

$$\boxed{r_{\mathrm{C}}(m) = \frac{\hbar}{mc\,\alpha}.} \tag{12}$$

This reproduces the reduced Bohr scale [1] as the hydrogenic realization of routing-suppressed closure.

4 Planck Energy Density Cap

Each lattice four-volume $\ell_{\mathrm{P}}^3 t_{\mathrm{P}}$ supports at most one radian of coherent phase evolution and therefore transports at most one quantum of reduced action $\hbar$ per tick. Because energy is action per unit time, the corresponding maximal local energy density is

$$\boxed{u_P \equiv \frac{\hbar}{\ell_{\mathrm{P}}^3 t_{\mathrm{P}}}.} \tag{13}$$

This bound is a *local* lattice constraint (an action-transport cap), not a global cutoff:

$$\boxed{u(r) \leq u_P.} \tag{14}$$

It is convenient to define the dimensionless occupancy fraction

$$\boxed{\eta \equiv \frac{u}{u_P}, \qquad 0 \leq \eta \leq 1.} \tag{15}$$

5 Saturation Impedance Blow-Up

To enforce the local cap dynamically, we adopt the minimal analytic divergence consistent with vanishing routing headroom at full occupancy $\eta = 1$. Specifically, we require that effective impedance remain finite for $\eta < 1$, diverge only at saturation, and reduce to the vacuum value at $\eta = 0$.

The simplest admissible functional form satisfying these conditions is

$$\boxed{\frac{Z_{\mathrm{eff}}(\eta)}{Z_{\mathrm{P}}} = (1-\eta)^{-1}.} \tag{16}$$

This linear pole is analytic for $\eta < 1$, diverges only at saturation, and guarantees that effective routing shuts off continuously as occupancy approaches unity.

Using the geometric routing admittance

$$\mathcal{Y} \equiv \frac{Y_{\mathrm{eff}}}{Y_{\mathrm{P}}},$$

the equivalent admittance suppression law becomes

$$\boxed{\mathcal{Y}(\eta) = (1 - \eta).} \tag{17}$$

As $\eta \to 1$,

$$Z_{\text{eff}} \to \infty, \qquad \mathcal{Y} \to 0, \tag{18}$$

so a fully saturated region becomes an infinite-impedance termination for additional phase–action transport.

6 Hard-Core Reflection

Consider phase–action transport across a radial boundary separating two regions of effective routing impedance. Let

- Z_1 denote the effective impedance of the exterior (unsaturated) region, and
- Z_2 denote the effective impedance of the interior region.

These impedances characterize resistance to phase–action transport in the same normalized sense introduced in Eq. (16): they describe routing availability rather than material properties.

For an impedance discontinuity, the reflection coefficient for an incoming phase–action disturbance is

$$\boxed{\mathcal{R} = \frac{Z_2 - Z_1}{Z_2 + Z_1}.} \tag{19}$$

In the present context:

- $Z_1 \sim Z_{\text{P}}$ for the exterior vacuum or partially occupied region ($\eta < 1$),
- $Z_2 = Z_{\text{eff}}(\eta)$ for the interior region.

Using the saturation law of Eq. (16),

$$\frac{Z_{\text{eff}}}{Z_{\text{P}}} = (1 - \eta)^{-1}, \tag{20}$$

the interior impedance diverges as $\eta \to 1$. Equivalently, from the geometric routing admittance defined in Eq. (3),

$$\mathcal{Y} = \frac{Y_{\text{eff}}}{Y_{\text{P}}},$$

one has $\mathcal{Y} \to 0$ as saturation is approached.

In the fully saturated limit,

$$Z_2 \to \infty, \tag{21}$$

and therefore

$$\mathcal{R} \to 1. \tag{22}$$

A fully saturated region thus behaves as an infinite-impedance termination for phase–action transport. No additional inward routing is dynamically admissible, and collapse cannot proceed beyond the saturated core boundary.

Worked example. To make Eq. (19) explicit, take the exterior to be unsaturated vacuum routing so that

$$Z_1 = Z_{\mathrm{P}}.$$

For a representative near-saturation occupancy, choose

$$\eta = 0.90.$$

Then the saturation law gives

$$\frac{Z_2}{Z_{\mathrm{P}}} = (1-\eta)^{-1} = (1-0.90)^{-1} = 10, \qquad \Rightarrow \qquad Z_2 = 10\, Z_{\mathrm{P}}. \tag{23}$$

Substituting into the reflection coefficient,

$$\mathcal{R} = \frac{Z_2 - Z_1}{Z_2 + Z_1} = \frac{10Z_{\mathrm{P}} - Z_{\mathrm{P}}}{10Z_{\mathrm{P}} + Z_{\mathrm{P}}} = \frac{9Z_{\mathrm{P}}}{11Z_{\mathrm{P}}} = \frac{9}{11} \approx 0.818. \tag{24}$$

As saturation increases, the impedance contrast grows. For example, if $\eta = 0.99$, then $(1-\eta)^{-1} = 100$ and

$$\mathcal{R} = \frac{100-1}{100+1} = \frac{99}{101} \approx 0.980. \tag{25}$$

In the fully saturated limit $\eta \to 1$, one has $Z_2/Z_{\mathrm{P}} \to \infty$, so

$$\mathcal{R} \to 1,$$

recovering perfect reflection as in Eq. (22).

7 Saturated Core Radius

The Planck energy-density cap implies a maximal local energy density

$$u_P = \frac{\hbar}{\ell_{\mathrm{P}}^3 t_{\mathrm{P}}}, \tag{26}$$

and therefore a maximal lattice mass density

$$\rho_{\max} = \frac{u_P}{c^2}. \tag{27}$$

This cap and its dynamical enforcement were established in [10]. Once the local occupancy reaches $\eta = 1$, further compression is dynamically forbidden.

Let r denote the general radial coordinate in the spacetime exterior. We now define R_c as the specific radius at which the interior first becomes fully saturated ($\eta = 1$). The minimal non-singular interior completion is therefore a uniform saturated core of radius R_c and density $\rho_{\max}$.

The total mass of such a core is the saturated density times its volume:

$$M = \rho_{\max} \frac{4\pi}{3} R_c^3. \tag{28}$$

Substituting (27) gives

$$M = \frac{u_P}{c^2} \frac{4\pi}{3} R_c^3. \tag{29}$$

Using the primitive identity

$$c = \frac{\ell_{\mathrm{P}}}{t_{\mathrm{P}}},$$

and the definition of the QLM Planck mass

$$m_P \equiv \frac{\hbar t_{\mathrm{P}}}{\ell_{\mathrm{P}}^2}, \tag{30}$$

one finds

$$\frac{u_P}{c^2} = \frac{\hbar}{\ell_{\mathrm{P}}^3 t_{\mathrm{P}}} \cdot \frac{t_{\mathrm{P}}^2}{\ell_{\mathrm{P}}^2} = \frac{m_P}{\ell_{\mathrm{P}}^3}.$$

Equation (29) therefore becomes

$$\boxed{M = m_P \frac{4\pi}{3} \left(\frac{R_c}{\ell_{\mathrm{P}}}\right)^3} \tag{31}$$

which is purely expressed in lattice primitives.

Solving for R_c yields

$$\boxed{R_c(M) = \left(\frac{3}{4\pi} \frac{M}{m_P}\right)^{1/3} \ell_{\mathrm{P}}.} \tag{32}$$

Gravitational collapse therefore terminates at a finite interior radius $R_c(M)$, set entirely by $\{\hbar, \ell_{\mathrm{P}}, t_{\mathrm{P}}\}$ and the universal density cap. The exterior geometry remains Schwarzschild, as discussed in [10].

8 Gravitational Throttling

Lorentz kinematics derived in [11] establish invariant lattice transport from per-radian Planck primitives and proper-tick accumulation. In the same QLM interpretation, a static gravitational field acts as an effective impedance that throttles the mapping between local proper-time update rates and rates measured relative to an observer at infinity.

For a spherically symmetric mass M, the Schwarzschild redshift factor may be written in primitive form as

$$\boxed{Z_g(r) = \left(1 - \frac{r_s}{r}\right)^{-1/2}, \qquad r_s = 2\left(\frac{M}{m_P}\right) \ell_{\mathrm{P}}.} \tag{33}$$

Here r denotes the exterior radial coordinate ($r \geq R_c$), and $Z_g(r)$ is not a modification of general relativity but a re-expression of the standard redshift factor as an impedance-like throttle on asymptotic transport.

Local action transport remains fixed by the lattice update rule, but the rate mapped to infinity is reduced by $Z_g(r)$:

$$\boxed{\left(\frac{dS}{dt}\right)_\infty = \frac{1}{Z_g(r)} \left(\frac{dS}{d\tau}\right).} \tag{34}$$

Because local invariants are unchanged while only the asymptotic mapping is reduced, the same factor applies to the Planck energy-density cap. The asymptotically referenced density transport is therefore

$$\boxed{u_\infty(r) = \frac{u_P}{Z_g(r)}.} \tag{35}$$

Since the exterior vacuum field depends only on the total enclosed mass (Birkhoff's theorem; see [4]), replacing the central singularity with a finite saturated core modifies only the interior completion and leaves the exterior Schwarzschild solution exactly unchanged.

9 Conclusion

This paper establishes the routing and saturation completion layer of the Quantum Lattice Model:

- The geometric routing admittance $\mathcal{Y}$ emerges as a normalized effective admittance ratio determined purely by boundary geometry.
- Closure radii follow directly from routing suppression.
- The Planck energy density cap enforces the local constraint $\eta \leq 1$.
- Minimal analytic impedance blow-up produces hard reflection at saturation.
- Gravitational collapse terminates at a finite core radius $R_c(M)$.
- By spherical symmetry and Birkhoff's theorem, the exterior field remains exactly Schwarzschild.

Gravity is therefore interpreted as asymptotic throttling of local phase–action transport, while collapse completion is governed by a universal lattice density bound. No new primitives, free parameters, or exterior modifications of general relativity are introduced.

The geometric routing admittance and saturated-core completion extend the QLM framework beyond Lorentz kinematics, providing a self-consistent geometric and dynamical completion of gravitational collapse within the reduced-action lattice structure.

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